2024 14th International Workshop on the Electromagnetic Compatibility of Integrated Circuits (EMC Compo) | 979-8-3315-0463-2/24/$31.00 ©2024 IEEE | DOI: 10.1109/EMCCompo61192.2024.10742020

A Simulation Workflow for Predicting IC Stripline Radiated Emissions of Bond Wire-Based Systems

Dominik Kreindl
ams OSRAM Group
Premstaetten, Austria
dominik.kreindl@ams-osram.com

Bernhard Weiss
ams OSRAM Group
Premstaetten, Austria
bernhard.weiss@ams-osram.com

Christian Stockreiter
ams OSRAM Group
Premstaetten, Austria
christian.stockreiter@ams-osram.com

Thomas Bauernfeind
Institute of Fundamentals and Theory in Electrical Engineering, TU Graz
Graz, Austria
t.bauernfeind@tugraz.at

Manfred Kaltenbacher
Institute of Fundamentals and Theory in Electrical Engineering, TU Graz
Graz, Austria
manfred.kaltenbacher@tugraz.at

***Abstract*—Light sources in optical sensor packages can cause high levels of electromagnetic interference due to their high current consumption and short switching times. The radiating structures are electrically short for the frequency range of interest for metrological verification according to the IEC 61967-8 standard. This can be used to find an equivalent representation of the emission sources in terms of Hertzian dipole moments. This paper presents a purely analytical simulation workflow for predicting radiated emissions in IC stripline measurements by employing dipole moments. The procedure is evaluated against finite element simulations and measurement data. The results show promising correlations in the lower frequency range, but significant deviations are observed above 1.5 GHz. Further research is needed to identify the root cause of these deviations.**

Keywords—finite element method, Hertzian dipole moment, IC stripline, radiated emissions.

I. Introduction

Vertical cavity surface emitting lasers (VCSELs) or light emitting diodes (LEDs) are used as a light source for various optical semiconductor applications. In many cases, one or more bond wires are required to conduct the operating current to the anode contact of the emitting die. Especially for VCSEL setups used, e.g., for proximity or direct time of flight (TOF) sensors, this general topology is currently unavoidable, e.g., in a stacked die configuration, as shown in Fig. 1. A setup like this can cause high electromagnetic interference (EMI) since the bond wire forms a loop, which will excite a magnetic Hertzian dipole moment [1], p.112. In certain orientations, this dipole moment will cause high inductive coupling to the IC stripline in an emission measurement according to the IEC 61967-8 standard [2]. If, in addition to the high coupling factors, a high and fast pulsed operating current on the bond wires is required, like in most VCSEL-based applications, setups like this can cause the risk of high results in the above-mentioned metrological procedures. Therefore, it is of high interest to have simulation workflows available to predict the emission measurement results to avoid expensive redesigns. Unfortunately, generating a precise EM model, including the complete in-silicon driver IC metal structures, requires much effort and would often be numerically too expensive, e.g. [3]. For this reason, abstracted and simplified yet accurate models are beneficial. In this paper, we present a purely analytical approach as well as a finite element method (FEM) based methodology for this prediction. Section II will present the underlying theory of the behavior of bond wires and magnetic dipole moments inside transversal electromagnetic (TEM) waveguides like IC striplines or TEM cells. In Section III, the simulation workflows mentioned above are introduced, and the results are discussed. The fourth and final section concludes the work and gives an outlook.

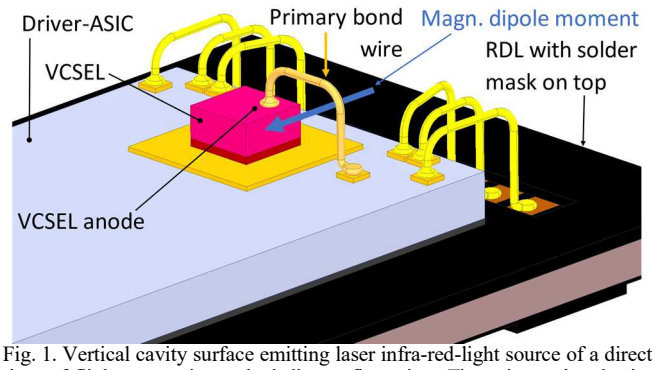

Fig. 1. Vertical cavity surface emitting laser infra-red-light source of a direct time of-flight-sensor in stacked die configuration. The primary bond wire connects the anode contact of the VCSEL with the driver ASIC and spans a current loop that generates a magnetic Hertzian dipole moment.

II. Theory and Fundamentals

The behavior of Hertzian dipole moments inside TEM-like waveguides is well-investigated and will be summarized in Section II-A). Based on this theory, the applicability to bond wire will be investigated based on measurements of a test PCB and FEM simulation.

A. Frequency response of Hertzian dipole moments to TEM waveguides

The behavior of infinitesimal line and loop currents inside waveguides is well described in [4], p. 358ff. Sreenivasiah et al. [5] used this work to develop a dipole extraction methodology from TEM cell measurements. The portability to IC striplines was investigated in [6]. The findings of these investigations can be summarized as follows:

The output voltages of the waveguide, when excited with correctly oriented electric and magnetic dipole moments, can be written as

$$\begin{pmatrix} U_1 \\ U_2 \end{pmatrix} = -\frac{Z_W}{2h}\left(m_{ez} \mp j\frac{m_{my}}{Z_0}\right). \quad (1)$$

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The magnitudes of the equivalent electric and magnetic dipole moments, extracted from voltage measurements at the waveguide output according to (1), are given to be:

$$|m_{ez}| = \frac{|U_1+U_2|h}{Z_W} \quad , \quad |m_{my}| = \frac{|U_1-U_2|hZ_0}{Z_W}. \qquad (2)$$

In (1) and (2), U_1 and U_2 are the output voltages of the waveguide (V), m_{ez} is an electric dipole moment in z-direction (Am), m_{my} is a magnetic dipole moment (Vm) in y-direction, h is the septum distance (m) , see Fig. 2, Z_W is the characteristic impedance of the waveguide, usually 50 Ω, Z_0 is the intrinsic impedance of free space, 120π Ω, and j is the imaginary unit. The dependency between the dipole intensity m_m and the geometry of an electrically small loop current, like a bond wire, is given by [7], p. 241:

$$|m_m| = |I_m l| = |j\Gamma\omega\mu_0 I_0|. \qquad (3)$$

In (3), I_0 is the loop line current, Γ is the area bounded by the line current, ω is the angular frequency, and μ_0 is the magnetic field constant of free space.

B. Dipole investigation on bond wire test PCB

To evaluate the applicability of the above presented theoretical model on bond wire structures, a 100 x 100 mm test PCB was built, featuring a single aluminum wire on the bottom side, bonded from a signal pad to ground, as depicted in Fig. 3 and Fig. 4. The signal pad is connected to a SMA connector soldered onto the top layer through a single via. The rest of the bottom and top layers are ground planes connected through appropriate via stitching. The PCB is put into an 8 mm IC stripline, and a three-port touchstone S-parameter file is extracted using a VNA. Additionally, an FEM model of the whole setup, including the PCB and the IC stripline, was generated and solved with a commercial frequency domain 3D full-wave solver. The measured and simulated touchstone files correlate well within 3 dB over the entire frequency range. In the next step, the extracted S-parameters are employed in a frequency domain schematic simulation, where a unity current of 1 A is impressed to port 1, and the voltages at port 2 and port 3 are monitored, with both ports terminated with 50 Ω (characteristic impedance of the used IC stripline). With the results, the equivalent dipole moments according to (2) are extracted. The magnetic dipole moment is also computed by employing (3), with the marked area in Fig. 3 used as Γ. The results are compared in Fig. 5. The comparison between the output voltages of the IC stripline in the schematic simulation and the purely analytical approach utilizing (1) and (3) can be seen in Fig. 6. Note that a deviation from the analytical 20 dB/decade slope is noticeable in the schematic simulation results in the higher frequency range. This is because the investigated current loop can no longer be considered electrically short in this frequency range, which is a prerequisite for using the analytical model. This is supported by Z11 of the used S-files (both measured and FEM-extracted, not shown in this paper due to space limitations). The curves for the real and imaginary parts of Z11 indicate a resonant behavior at 1.9 GHz. For smaller bond wire structures (like in actual IC packages), this resonant behavior occurs at higher frequencies. It, therefore, should not be a problem for the target application of the proposed analytical approach. This is

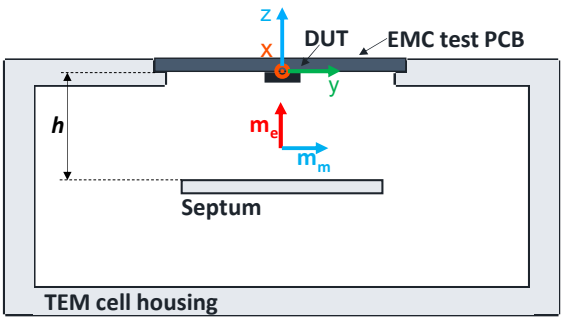
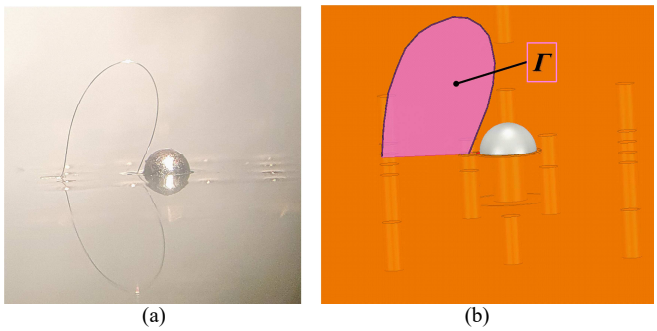


Fig. 2. Cross-section of a TEM cell with an electric (z-axis) and a magnetic Hertzian dipole moment (y-axis). Aligned this way, the dipole moments will generate a propagating mode within the waveguide, contributing to the output voltages of the TEM cell according to (1).



(a) (b)

Fig. 3. The 25 um bond wire geometry of the test PCB described in Section II-B. (a) Photograph of the test structure. (b) FEM model of the test structure with the area Γ used for the analytical dipole calculation highlighted; the wire is colored black for better visibility. The small sphere is a solder ball.

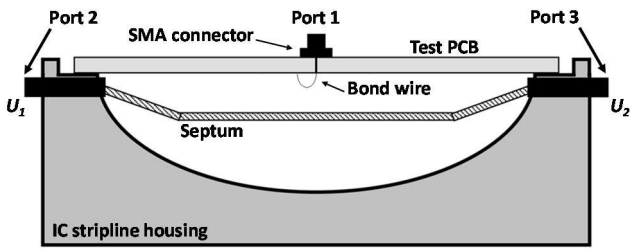


Fig. 4. Measurement and simulation setup for the three-port S-parameter extraction: The test PCB is mounted to the IC stripline oriented, so the magnetic Hertzian dipole moment generated by the bond wire current loop is perpendicular to the septum (see Fig. 2).

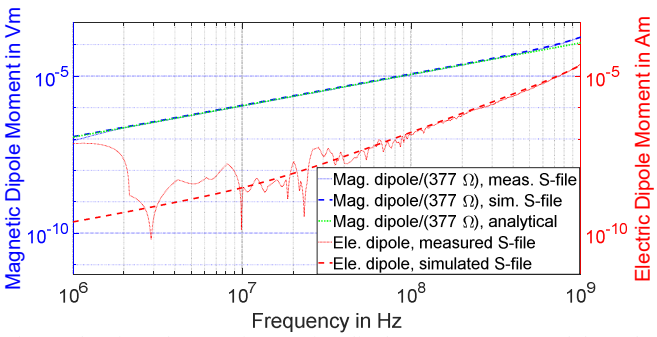


Fig. 5. Electric and magnetic Hertzian dipole moments extracted from the simulated and measured three-port S-parameter files and the purely analytically calculated magnetic Hertzian dipole moment. The magnetic moments are normalized by the intrinsic impedance of free space, so they are comparable to the electric moments in the same diagram in how they contribute to the output voltages of the TEM cell or IC stripline according to (1).

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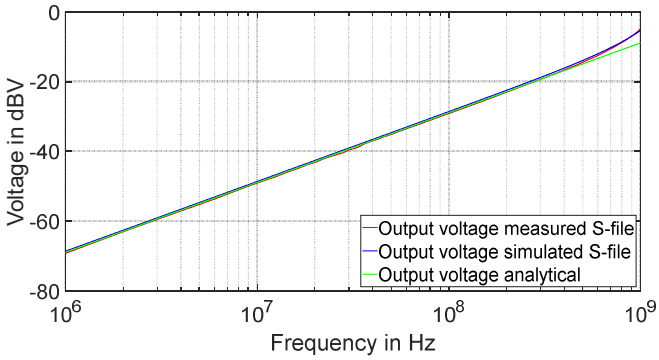

Fig. 6. Output-voltages of the IC stripline in the setup of Fig. 4 when port 1 is excited with 1 A for the measured, and FEM simulated S-parameter file as well as calculated purely analytical.

also supported by experiments with test PCBs featuring shorter bond wire structures. For those smaller structures, it becomes difficult to measure the low coupling factors between the bond wire and the IC stripline S21 and S31 in the lower frequency (LF) range with a VNA. Fig. 5 also shows that the equivalent electric dipole moment of a bond wire current loop is insignificant compared to the magnetic dipole moment. Therefore, the electric coupling will not be considered for the workflow presented in the next section.

III. Workflow

Based on the promising results of the previous section, the methodology is evaluated on an actual VCSEL product.

A. General idea

The geometry of the bond wire and stacked die structure is known and depicted in Fig. 1. Therefore, the frequency response to the IC stripline can be approximated by (1) and (3) in the same way as before. The used area for Γ is shown in Fig. 7. Of course, in an actual application, the current loop is not driven with a frequency-constant current but with the pulsed operating current of the VCSEL. This current pulse is known from time-domain functional schematic simulations of the device. The device-specific pulse shape is shown in blue in Fig. 8. This pulse is repeated with a pulse frequency of 17.75 MHz during the distance measurement of the TOF sensor. To load the loop with the actual operating current in the frequency domain, a fast Fourier transformation (FFT) is performed on the signal. The single-sided spectrum is shown in Fig. 9. Since the complete system is expected to be linear, every frequency point can be treated separately. Applying this to the problem at hand, the workflow can be summarized as follows:

- Find the frequency response of the current loop to the IC stripline or TEM cell using (1) and (3).
- Apply FFT on time domain (TD) operating current (available from analog design schematic simulations).
- Multiply the frequency-domain signals to get the final emission spectrum (allowed since the complete system is considered to be linear).

This workflow procedure is depicted in Fig. 10.

The measured emission spectrum of the VCSEL product at hand according to the standard [2] is shown in Fig. 11. Note that only harmonics of the VCSEL current (multiples of

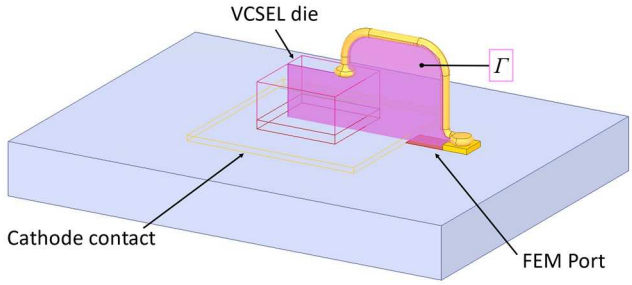

Fig. 7. FEM model of the VCSEL light source of the investigated TOF sensor. The area Γ used for the analytical workflow is highlighted in magenta.

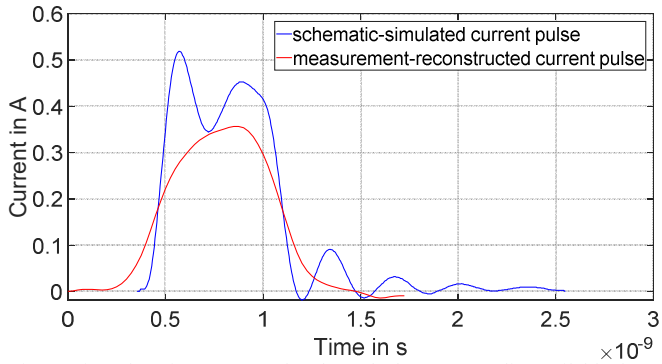

Fig. 8. Time-domain current pulse through the VCSEL die available from schematic analog design functional simulations and reconstructed by TD IC stripline voltage measurements.

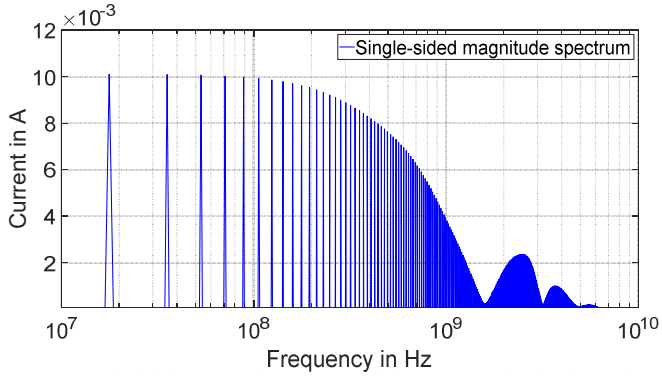

Fig. 9. Single-sided magnitude spectrum of the current pulse in Fig. 8 when the pulse is repeated with a fundamental pulse frequency of 17.75 MHz. Note: The y-axis is scaled linearly.

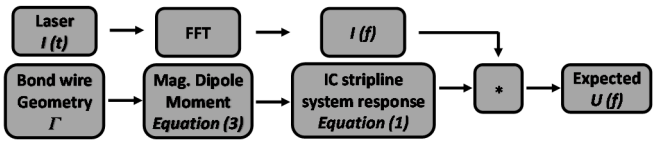

Fig. 10. Proposed purely analytical workflow methodology for predicting the voltage levels in IC stripline RE measurements of bond wire-based systems.

17.75 MHz) are considered for the envelope building. The emission spikes whose frequencies are not multiples of the fundamental frequency are not caused by the investigated bond wire and are therefore ignored for the envelop-building, as these emissions are not within the scope of this work.

The envelope of Fig. 11 is compared to the result of the proposed simulation workflow (blue curve) in Fig. 12. The curves show good agreement in the LF range. In the upper frequency range, a significant derivation is observable. Since the procedure works well for the test PCB from Section II and

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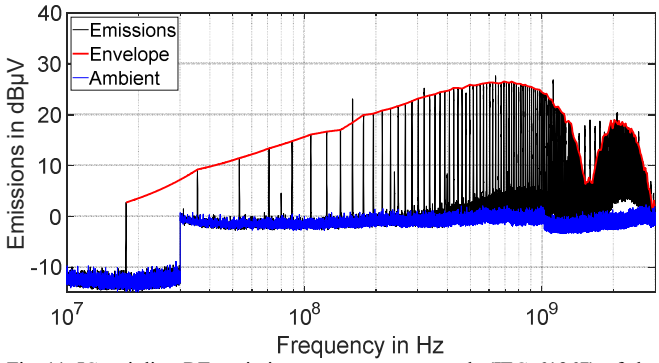

Fig. 11. IC stripline RE emissions measurement result (IEC 61967) of the investigated TOF sensor. Note that emission spikes that are not caused by the primary VCSEL bond wire are ignored for the envelope building.

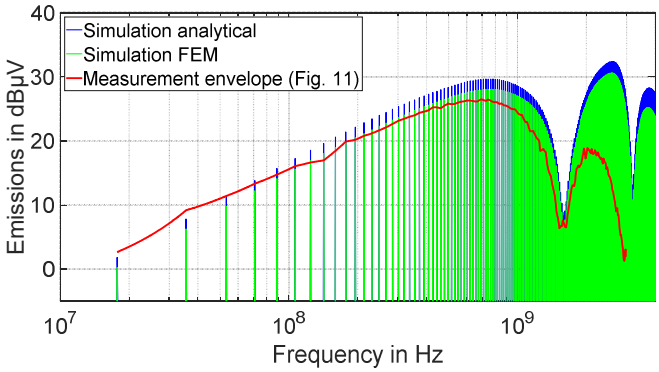

Fig. 12. Comparison between the measurement results envelope and the results of the proposed workflow, both purely analytical calculation as well as FEM extracted system frequency response.

for the LF range of the investigated sensor device, we will try to narrow down the source of error for the higher frequency range in Fig. 12 in the next section.

B. FEM model for frequency response

Now, instead of the analytical approach to get the frequency response from the current loop to the IC stripline, a FEM model of the loop, including the VCSEL and the IC stripline, is used for this extraction, as shown in Fig. 7, and similar to [8]. This way, capacitive couplings (i.e., electric dipole moments), proximity effects within the current loop or the non-ideal TEM properties of the IC stripline (reflections, higher modes, etc.), can be considered. All these effects are neglected in the proposed purely analytical workflow.

In the EM model in Fig. 7, the VCSEL was modeled as a bulk material with a conductivity of approximately 2000 S/m. The upper area is a perfectly conducting boundary condition. The excitation port was placed between the cathode area and the bond wire pad. This 3D VCSEL model is inserted into the 3D EM model of the IC stripline used in Section II, resulting in a three-port FEM model. The rest of the simulation procedure for the final emission spectrum is the same as before, as shown in Fig. 10. The results (in green) are compared to the analytical workflow (blue), and the envelope of the measurement results (red) in Fig. 12.

The authors are aware that the FEM model used is very simplified and lacks a realistic representation of the in-silicon current paths within the driver ASIC. Generating a more realistic model that can accurately represent the current return path in the ASIC would require a simplification and import workflow for the driver layout, which is beyond the scope of this work. Besides that, using a more realistic model of the current return path would not change the results significantly, since a return path more spread out over the area of the ASIC (x-y plane) would not add to the spanned current loop area of the bond wire in the x-z plane and therefore would not couple to the IC stripline. Horizontal current loops that do not couple to a TEM cell (and therefore do not couple to an IC stripline) have already been identified as a weakness of these characterization systems by Fiori et al. [9].

C. Discussion

In the LF range, both the analytical simulation approach (Section III-A) and the FEM-based approach (Section III-B) show a good correlation with the IC stripline radiated emission measurement. Above 1.5 GHz, both approaches show a significant deviation from the measurements. Since both methods show a similar deviation, it can be assumed that the source of the deviation lies in the simulation-extracted current pulse (Fig. 8, blue curve and Fig. 9). This current is extracted from functional schematic simulations. The main goal of these simulations is to verify the functionality of the ASIC and do not have the ambition to be exactly precise for higher frequencies above 1 GHz, since parasitic effects occurring in this frequency range would need much more effort to be accurately considered. However, the pulse shape was used in this research because the goal was also to enable the proposed simulation workflow with data already available from the product development process.

To support this, a TD measurement of the IC stripline output voltage was performed using an oscilloscope with 6 GHz bandwidth and 40 GS/s sampling rate. This measurement is necessary to obtain the full signal information because the used EMI receiver is not able to extract the phase shift information between the individual frequency components, and therefore the TD voltage pulse cannot be reconstructed by post-processing (i.e., inverse discrete Fourier transform) the standardized RE measurement. Using the oscilloscope-measured voltage pulse (Fig. 13), the proposed workflow was reversed to extract the TD shape of the current pulse on the bond wire. To do this, (1) and (3) are inverse Fourier transformed and rearranged (neglecting m_{ez}) to

$$I_0 = \pm \frac{2hZ_0}{\Gamma\mu_0 Z_W} \int \begin{pmatrix} U_1 \\ U_2 \end{pmatrix} dt. \tag{4}$$

The result for the measurement-reconstructed current pulse I_0 is shown in Fig. 8 (red curve). Comparing it to the originally used pulse shape (blue curve), suggests that the inductance of the current path was underestimated in the aforementioned schematic simulations.

IV. Conclusion and Outlook

This paper presents a purely analytical methodology for predicting standardized radiated emissions characterization results of bond wire-based systems, using only data already available from the product development process. The method is evaluated by means of numerical field simulations and VNA and EMI receiver measurements. The computed emission magnitude curves show a very promising correlation

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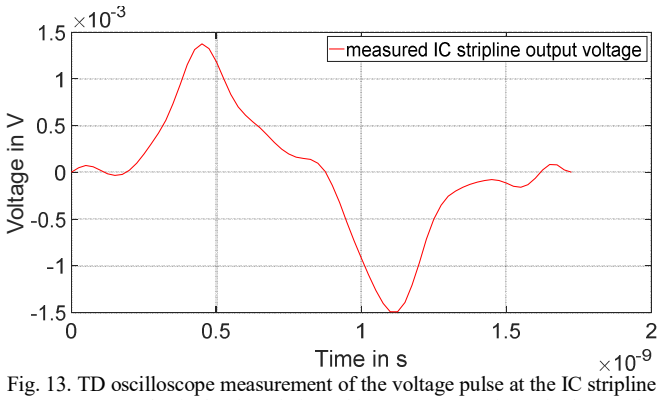

Fig. 13. TD oscilloscope measurement of the voltage pulse at the IC stripline output. From a physical point of view, this measurement is equivalent to the EMI receiver measurement shown in Fig. 11, the difference being the signal representation (TD vs. FD, Oscilloscope vs. EMI receiver).

to the measurements in the frequency range below 1.5 GHz. The significant deviations above 1.5 GHz are suspected to be caused by insecurities in the current pulse used. A reconstruction of the current pulse based on TD oscilloscope measurements supports this. Investigations to improve the schematic simulation-based determination of this pulse are part of ongoing research activities. A straightforward starting point could be to replace purely schematic models for the VCSEL load in the current pulse simulation with FEM-extracted S-parameter models, since parasitic effects are more accurately considered. To do so, the development of improved FEM VCSEL models is necessary for future research, too.

ACKNOWLEDGMENT

This research is funded by ams OSRAM.

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